1) $53:5 = 10\ (3)$

$10:5 = 2\ (0)$

$\boxed{53_{10} = 203_5}$

$53:16 = 3\ (5) = \boxed{35}$

$3:16 =$

$53_{10} = 35h$

$53:8 = 6\ (5) = \boxed{65}$

$\boxed{53_{10} = 65_8}$

$53:2 = 26\ (1)$

$26:2 = 13\ (0)$

$13:2 = 6\ (1)$

$6:2 = 3\ (0)$

$3:2 = 1\ (1)$

$\boxed{53_{10} = 110101_2}$

2) $203_5 = 3\cdot 5^0 + 0\cdot 5^1 + 2\cdot 5^2 = 53$

$35h = 5\cdot 16^0 + 3\cdot 16^1 = \boxed{53}$

$65_8 = 5\cdot 8^0 + 6\cdot 8^1 = \boxed{53}$

$110101_2 = 1 + 2^2 + 2^4 + 2^5 = \boxed{53}$

5)

1) $11100101_2 = 1 + 2^2 + 2^5 + 2^6 + 2^7 = 229_{10} = e5h$

2) $01101001_2 = 1 + 2^3 + 2^5 + 2^6 = 105_{10} = 69h$

3) $10010110_2 = 2 + 2^2 + 2^4 + 2^7 = 150_{10} = 96h$

4) $10000000 = 2^7 = 128_{10} = 80h$

5) $01111111 = 127_{10} = f7h$

1) 11100101
0001 1010
0001 1011 $= \boxed{-20}$

4) 1000 0000
0111 1111
1000 0000 $= \boxed{-128}$

3) 10010110
01101001
01101010 $= \boxed{-106}$

6) $_k$ -51_{10} =

51 : 2 = 25(1)
25 : 2 = 12(1)
12 : 2 = 6(0)
6 : 3 = 3(0)
3 : 2 = 1(1)

= 110011

00110011
11001100
11001101

7) -4_{10} = 4 : 2 = 2(0)
2 : 2 = 1(0) = 100

000001000
11111011
11111000 = -4

8) -128_{10} = 10000000
01111111
10000000 = -128_{10}

10) 0000 0000 0011 0011
1111 1111 1100 1100
1111 1111 1100 1101

9) 0000000000000001000
1111111111111111011
1111 1111 1111 1111 1000

8) 0000 0000 10000000
1111 1111 0111 1111
1111 1111 1000 0000

---

6) $_k$ -1_{10} = 1111 1111 1111 1111

9) -21400_{10} = 10111000 11 001000

-21400 : 2 = 10700(0)
10700 : 2 = 5350(0)
5350 : 2 = 2675(0)
2675 : 2 = 1337(1)
1337 : 2 = 668(1)
668 : 2 = 334(0)
334 : 2 = 167(0)
167 : 2 = 83(1)
83 : 2 = 41(1)
41 : 2 = 20(1)
20 : 2 = 10(0)
10 : 2 = 5(0)
5 : 2 = 2(1)
2 : 2 = 1(0)

10100111 00 111000
01011000 11 000111

8) -1024_{10} = 1024 : 2 = 512(0)$_1$ 1111110000000000
512 : 2 = 256(0)$_2$
256 : 2 = 128(0)$_3$
128 : 2 = 64(0)$_4$
64 : 2 = 32(0)$_5$
32 : 2 = 16(0)$_6$
16 : 2 = 8(0)$_7$
8 : 2 = 4(0)$_8$
4 : 2 = 2(0)$_9$
2 : 2 = 1(0)$_{10}$

0000100000000000
1111011111111111

7) k)

```
  11110001
+ 00010010
  11110011
```

ZF=0 CF=0
SF=1 OF=0

2)

```
  11111111
+ 11111111
(1)11111110
```

ZF=0 CF=1
SF=1 OF=0

r)

```
  01111111
+ 01111111
  11111110
```

ZF=0 CF=0
SF=1 OF=1

3)

```
  01101011
+ 11101110
(1)01011001
```

ZF=0 CF=1
SF=0 OF=0

7)

```
(1)11111110
   0000 0000
 - 0000 0001
   1111 1111
```

ZF=0 CF=0
SF=1 OF=1

1)

```
   0110 0100
 - 0000 1111
   0101 0101
```

ZF=0 CF=0
SF=0 OF=0

8)

```
  1000
+ 1A2C
  2A2C
```

```
  4BF2
+ 362C
  7A2E
```

```
  2abd
+ 3105
  5bC2
```

9)

$68_{10} = 2^6 + 2^2 = 0100\ 0100$

$1029 = 2^{10} + 2^2 + 1 = 10000000101$

$520 = 2^9 + 2^3 = 1000001000$

$32770 = 2^{15} + 2 = 1000\ 0000\ 00000010$